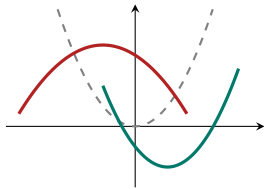


Prof. Fang Math

Experience first. Symbols second. Understanding always.

TEACHER VERSION

Name: _____



Today's Learning Targets

I can ...

- recognize the **parent** quadratic function and its graph
- describe how a , h , and k **transform** the graph
- find the vertex, axis of symmetry, max/min, domain and range from **vertex form**
- **write** a quadratic function from a graph or a description

Vocabulary

parabola	vertex form
vertex	minimum value
axis of symmetry	maximum value
reflection	translation
vertical stretch	vertical compression

Teacher: Read the targets aloud. Ask: "Where have you seen a U-shape like this before?" (fountains, basketball shots, bridges).

Warm-Up: The Parent Function $f(x) = x^2$

Complete the table, then plot the points.

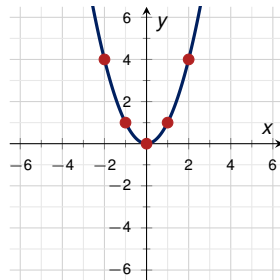
x	-3	-2	-1	0	1	2	3
$f(x)$	—	—	—	—	—	—	—

Notice

What do you notice about the outputs for x and $-x$?

They are _____, so the graph is
_____.

Teacher: Experience first: let students plot before naming anything. Point to the lowest point and the mirror line.



Warm-Up: The Parent Function $f(x) = x^2$

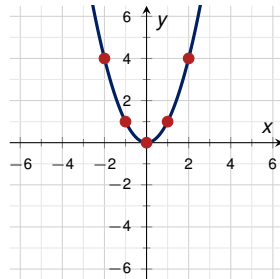
Complete the table, then plot the points.

x	-3	-2	-1	0	1	2	3
$f(x)$	<u>9</u>	<u>4</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>4</u>	<u>9</u>

Notice

What do you notice about the outputs for x and $-x$?

They are the same, so the graph is symmetric.



Teacher: Experience first: let students plot before naming anything. Point to the lowest point and the mirror line.

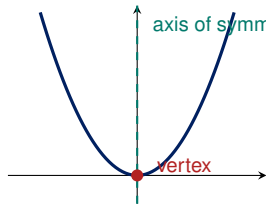
Definition: Quadratic Function

Key Idea

A **quadratic function** is a function that can be written as $f(x) = ax^2 + bx + c$, where _____.

Its graph is a U-shaped curve called a _____.

- The **vertex** is the _____ of the parabola.
- The **axis of symmetry** is the _____ through the vertex.
- For $f(x) = x^2$: vertex _____, axis _____



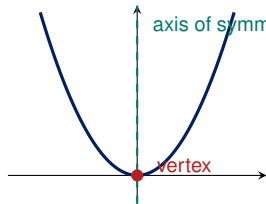
Definition: Quadratic Function

Key Idea

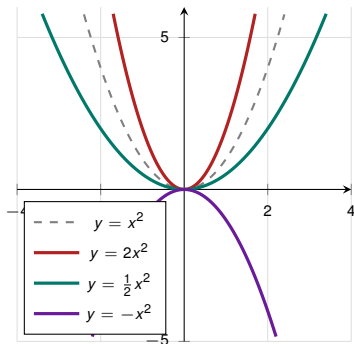
A **quadratic function** is a function that can be written as $f(x) = ax^2 + bx + c$, where $a \neq 0$.

Its graph is a U-shaped curve called a **parabola**.

- The **vertex** is the **turning point** of the parabola.
- The **axis of symmetry** is the **vertical line** through the vertex.
- For $f(x) = x^2$: vertex $(0, 0)$, axis $x = 0$



Explore 1: What Does a Do?

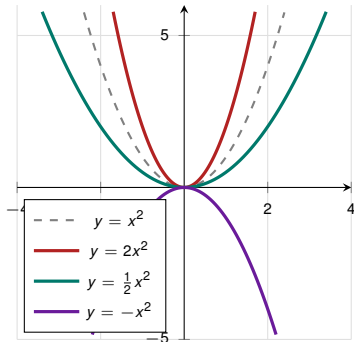


Notice

- $y = 2x^2$ is _____ than $y = x^2$.
(each y is _____)
- $y = \frac{1}{2}x^2$ is _____ than $y = x^2$.
- $y = -x^2$ opens _____: it is a
_____ in the x -axis.
- The vertex stays at _____ every time.

Teacher: Ask: "At $x = 1$, what are the heights of each graph?" $(1, 2, \frac{1}{2}, -1)$. Multiplying every output by a stretches the graph vertically.

Explore 1: What Does a Do?



Notice

- $y = 2x^2$ is narrower than $y = x^2$.
(each y is doubled)
- $y = \frac{1}{2}x^2$ is wider than $y = x^2$.
- $y = -x^2$ opens down: it is a reflection in the x -axis.
- The vertex stays at (0,0) every time.

Teacher: Ask: "At $x = 1$, what are the heights of each graph?" $(1, 2, \frac{1}{2}, -1)$. Multiplying every output by a stretches the graph vertically.

Key Idea: The Graph of $f(x) = ax^2$

Value of a	Effect on the graph	Example
$ a > 1$	vertical _____ (narrower)	$y = 3x^2$
$0 < a < 1$	vertical _____ (wider)	$y = \frac{1}{4}x^2$
$a > 0$	opens _____; vertex is a _____	$y = 2x^2$
$a < 0$	opens _____; vertex is a _____	$y = -2x^2$

Common Mistake

“Stretch” and “compression” depend on $|a|$, not on the sign: $y = -3x^2$ is a reflection *and* a vertical _____.

Key Idea: The Graph of $f(x) = ax^2$

Value of a	Effect on the graph	Example
$ a > 1$	vertical <u>stretch</u> (narrower)	$y = 3x^2$
$0 < a < 1$	vertical <u>compression</u> (wider)	$y = \frac{1}{4}x^2$
$a > 0$	opens <u>up</u> ; vertex is a <u>minimum</u>	$y = 2x^2$
$a < 0$	opens <u>down</u> ; vertex is a <u>maximum</u>	$y = -2x^2$

Common Mistake

“Stretch” and “compression” depend on $|a|$, not on the sign: $y = -3x^2$ is a reflection *and* a vertical stretch.

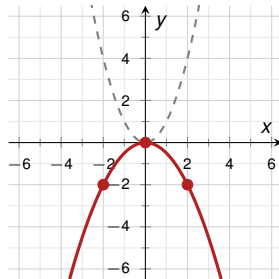
Watch Me 1: Graph $f(x) = -\frac{1}{2}x^2$

Watch Me

x	-4	-2	0	2	4
$f(x)$	___	___	___	___	___

Compared with $y = x^2$:

- reflection in the _____
- vertical _____ by a factor of _____
- vertex _____ is a _____



Teacher: Choose x -values that make $\frac{1}{2}x^2$ a whole number (even numbers).

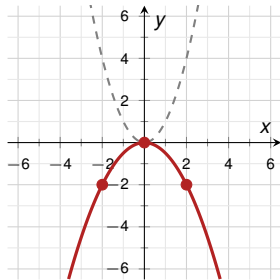
Watch Me 1: Graph $f(x) = -\frac{1}{2}x^2$

Watch Me

x	-4	-2	0	2	4
$f(x)$	<u>-8</u>	<u>-2</u>	<u>0</u>	<u>-2</u>	<u>-8</u>

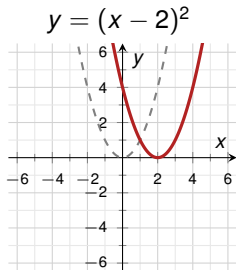
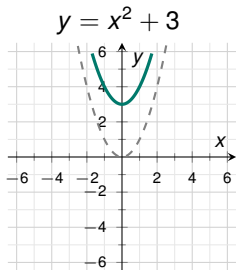
Compared with $y = x^2$:

- reflection in the **x-axis**
- vertical **compression** by a factor of $\frac{1}{2}$
- vertex **(0,0)** is a **maximum**



Teacher: Choose x -values that make $\frac{1}{2}x^2$ a whole number (even numbers).

Explore 2: What Do h and k Do?



Notice

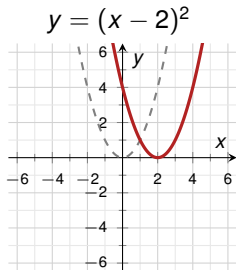
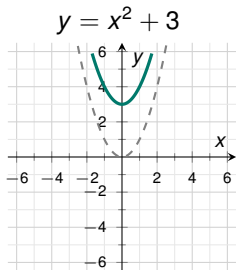
$y = x^2 + 3$ moves the graph
_____.

$y = (x - 2)^2$ moves the
graph _____.

Surprise? $x - 2$ *inside* the
square moves the graph
the _____ way of the
sign.

Teacher: Ask: "For $y = (x - 2)^2$, what x makes the output 0?" ($x = 2$) — so the vertex moved to $x = 2$, to the right.

Explore 2: What Do h and k Do?



Notice

$y = x^2 + 3$ moves the graph
up 3.

$y = (x - 2)^2$ moves the
graph right 2.

Surprise? $x - 2$ *inside* the
square moves the graph
the opposite way of the
sign.

Teacher: Ask: "For $y = (x - 2)^2$, what x makes the output 0?" ($x = 2$) — so the vertex moved to $x = 2$, to the right.

Key Idea: Translations

Function	Translation of $y = x^2$	Vertex
$y = x^2 + k$	_____ if $k > 0$, _____ if $k < 0$	_____
$y = (x - h)^2$	_____ if $h > 0$, _____ if $h < 0$	_____
$y = (x + 3)^2$	_____ (here $h =$ _____)	_____

Common Mistake

$(x + 3)^2$ is $(x - (-3))^2$, so $h = -3$: the graph moves **left**, not right.

Key Idea: Translations

Function	Translation of $y = x^2$	Vertex
$y = x^2 + k$	<u>up</u> if $k > 0$, <u>down</u> if $k < 0$	<u>$(0, k)$</u>
$y = (x - h)^2$	<u>right</u> if $h > 0$, <u>left</u> if $h < 0$	<u>$(h, 0)$</u>
$y = (x + 3)^2$	<u>left</u> <u>3</u> (here $h = \underline{-3}$)	<u>$(-3, 0)$</u>

Common Mistake

$(x + 3)^2$ is $(x - (-3))^2$, so $h = -3$: the graph moves **left**, not right.

Putting It Together: Vertex Form

Vertex Form

$$f(x) = a(x - h)^2 + k$$

- vertex: _____
- axis of symmetry: _____
- $a > 0$: opens up, min value _____
- $a < 0$: opens down, max value _____
- domain: _____
- range ($a > 0$): _____
- range ($a < 0$): _____
- $|a|$ controls _____

Order to graph

_____/stretch with $a \rightarrow$ shift _____ by $h \rightarrow$ shift _____ by k

Putting It Together: Vertex Form

Vertex Form

$$f(x) = a(x - h)^2 + k$$

- vertex: (h, k)
- axis of symmetry: $x = h$
- $a > 0$: opens up, min value k
- $a < 0$: opens down, max value k
- domain: **all real numbers**
- range ($a > 0$): $y \geq k$
- range ($a < 0$): $y \leq k$
- $|a|$ controls **stretch / compression**

Order to graph

reflect / stretch with $a \rightarrow$ shift **horizontally** by $h \rightarrow$ shift **vertically** by k

Watch Me 2: $f(x) = (x + 3)^2 - 4$

Watch Me

Rewrite: $f(x) = 1(x - (\underline{\hspace{1cm}}))^2 + (\underline{\hspace{1cm}})$

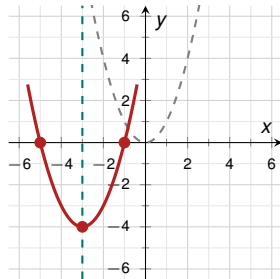
$a = \underline{\hspace{1cm}}$, $h = \underline{\hspace{1cm}}$, $k = \underline{\hspace{1cm}}$

Transformations: $\underline{\hspace{1cm}}$ 3, $\underline{\hspace{1cm}}$ 4

Vertex $\underline{\hspace{1cm}}$ Axis $\underline{\hspace{1cm}}$

$\underline{\hspace{1cm}}$ value: $\underline{\hspace{1cm}}$

Domain: $\underline{\hspace{2cm}}$ Range: $\underline{\hspace{2cm}}$



Teacher: Plot the vertex first, then use the parent pattern “over 1, up 1; over 2, up 4.”

Watch Me 2: $f(x) = (x + 3)^2 - 4$

Watch Me

Rewrite: $f(x) = 1(x - (\underline{-3}))^2 + (\underline{-4})$

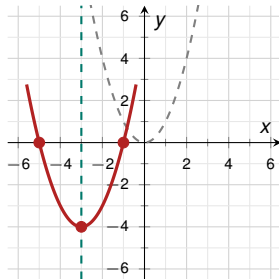
$a = \underline{1}$, $h = \underline{-3}$, $k = \underline{-4}$

Transformations: left 3, down 4

Vertex $(-3, -4)$ Axis $x = -3$

Minimum value: -4

Domain: all real numbers Range: $y \geq -4$



Teacher: Plot the vertex first, then use the parent pattern “over 1, up 1; over 2, up 4.”

Watch Me 3: $f(x) = -2(x - 1)^2 + 5$

Watch Me

$a = \underline{\hspace{1cm}}$, $h = \underline{\hspace{1cm}}$, $k = \underline{\hspace{1cm}}$

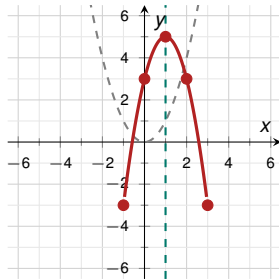
Transformations (in order):

$\underline{\hspace{1cm}}$, vertical $\underline{\hspace{1cm}}$ by $\underline{\hspace{1cm}}$,
 $\underline{\hspace{1cm}}$ 1, $\underline{\hspace{1cm}}$ 5

Vertex $\underline{\hspace{1cm}}$ Axis $\underline{\hspace{1cm}}$

$\underline{\hspace{1cm}}$ value: $\underline{\hspace{1cm}}$ Range: $\underline{\hspace{1cm}}$

Teacher: With $a = -2$ the pattern becomes “over 1, down 2; over 2, down 8.”



Watch Me 3: $f(x) = -2(x - 1)^2 + 5$

Watch Me

$a = \underline{-2}$, $h = \underline{1}$, $k = \underline{5}$

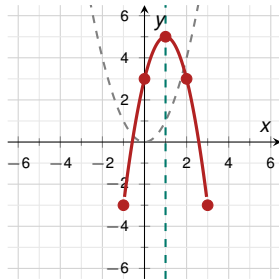
Transformations (in order):

reflect in x-axis, vertical **stretch** by **2**,
right 1, **up** 5

Vertex **$(1, 5)$** Axis **$x = 1$**

Maximum value: **5** Range: **$y \leq 5$**

Teacher: With $a = -2$ the pattern becomes “over 1, down 2; over 2, down 8.”



Your Turn: Read the Vertex Form

Your Turn

Function	Vertex	Axis	Max or Min?	Value
$f(x) = (x - 5)^2 + 2$	_____	_____	_____	_____
$g(x) = -(x + 1)^2 - 3$	_____	_____	_____	_____
$p(x) = \frac{1}{2}x^2 - 6$	_____	_____	_____	_____

Teacher: Give 2 minutes. Ask a student to explain why g has a maximum (it opens down because $a = -1$).

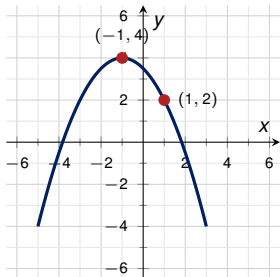
Your Turn: Read the Vertex Form

Your Turn

Function	Vertex	Axis	Max or Min?	Value
$f(x) = (x - 5)^2 + 2$	<u>(5, 2)</u>	<u>$x = 5$</u>	<u>min</u>	<u>2</u>
$g(x) = -(x + 1)^2 - 3$	<u>(-1, -3)</u>	<u>$x = -1$</u>	<u>max</u>	<u>-3</u>
$p(x) = \frac{1}{2}x^2 - 6$	<u>(0, -6)</u>	<u>$x = 0$</u>	<u>min</u>	<u>-6</u>

Teacher: Give 2 minutes. Ask a student to explain why g has a maximum (it opens down because $a = -1$).

Watch Me 4: Write the Equation from a Graph



Watch Me

Step 1. Vertex $(h, k) =$ _____

$$f(x) = a(x + ___)^2 + ___$$

Step 2. Use another point $(1, 2)$ to find a :

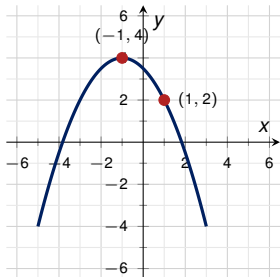
$$2 = a(1 + 1)^2 + 4$$

$$2 = ______ \Rightarrow a = ______$$

Step 3. $f(x) =$ _____

Check: opens _____, so a should be _____ ✓

Watch Me 4: Write the Equation from a Graph



Watch Me

Step 1. Vertex $(h, k) = \underline{(-1, 4)}$

$$f(x) = a(x + \underline{1})^2 + \underline{4}$$

Step 2. Use another point $(1, 2)$ to find a :

$$2 = a(1 + 1)^2 + 4$$

$$2 = \underline{4a + 4} \Rightarrow a = \underline{-\frac{1}{2}}$$

Step 3. $f(x) = \underline{-\frac{1}{2}(x + 1)^2 + 4}$

Check: opens down, so a should be negative ✓

Watch Me 5: Write the Equation from a Description

Watch Me

Start with $y = x^2$. Reflect it in the x -axis, stretch it vertically by a factor of 3, then translate it **4 units left** and **1 unit down**.

reflect + stretch by 3 $\Rightarrow a = \underline{\hspace{2cm}}$

4 units left $\Rightarrow h = \underline{\hspace{2cm}}$

1 unit down $\Rightarrow k = \underline{\hspace{2cm}}$

$g(x) = \underline{\hspace{4cm}}$

Common Mistake

Left 4 means $h = -4$, which appears as $(x + 4)$ in the equation.

Watch Me 5: Write the Equation from a Description

Watch Me

Start with $y = x^2$. Reflect it in the x -axis, stretch it vertically by a factor of 3, then translate it **4 units left** and **1 unit down**.

$$\text{reflect + stretch by 3} \Rightarrow a = \underline{-3}$$

$$4 \text{ units left} \Rightarrow h = \underline{-4}$$

$$1 \text{ unit down} \Rightarrow k = \underline{-1}$$

$$g(x) = \underline{-3(x + 4)^2 - 1}$$

Common Mistake

Left 4 means $h = -4$, which appears as $(x + 4)$ in the equation.

What If ... ?

What If?

1. What if a changes from 2 to -2 — does the vertex move?

2. What if you swap h and k : $(x - 2)^2 + 5$ vs. $(x - 5)^2 + 2$?

3. What if $a = 0$? Is $y = 0(x - 1)^2 + 3$ still a parabola?

Teacher: Pair-share first, then take answers. Item 3 explains why the definition needs $a \neq 0$.

What If ... ?

What If?

1. What if a changes from 2 to -2 — does the vertex move?

No. Only the direction changes.

2. What if you swap h and k : $(x - 2)^2 + 5$ vs. $(x - 5)^2 + 2$?

Different vertices: $(2, 5)$ vs. $(5, 2)$.

3. What if $a = 0$? Is $y = 0(x - 1)^2 + 3$ still a parabola?

No — it is the line $y = 3$.

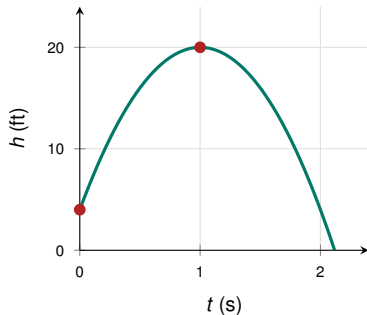
Teacher: Pair-share first, then take answers. Item 3 explains why the definition needs $a \neq 0$.

Connection: A Water Fountain

The height (feet) of a water stream t seconds after it leaves the fountain is

$$h(t) = -16(t - 1)^2 + 20.$$

1. Maximum height: _____
2. It reaches the top after _____.
3. Starting height $h(0) =$ _____
4. Why is the vertex a maximum?



Teacher: Domain here is only the time the water is in the air. Ask: “Does a negative t make sense?”

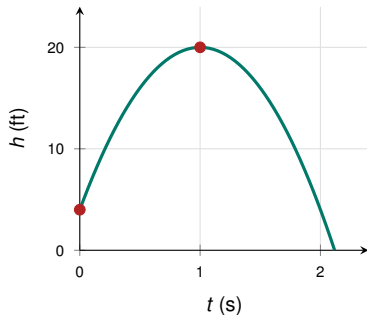
Connection: A Water Fountain

The height (feet) of a water stream t seconds after it leaves the fountain is

$$h(t) = -16(t - 1)^2 + 20.$$

1. Maximum height: 20 ft
2. It reaches the top after 1 second.
3. Starting height $h(0) =$ 4 ft
4. Why is the vertex a maximum?
 $a = -16 < 0$, so it opens down.

Teacher: Domain here is only the time the water is in the air. Ask: "Does a negative t make sense?"



Your Turn

1. Describe the transformations of $y = x^2$: $g(x) = 3(x - 2)^2 + 1$

2. Describe the transformations of $y = x^2$: $g(x) = -\frac{1}{4}(x + 5)^2$

3. Write the function: $y = x^2$ reflected in the x -axis, then 6 right and 2 up.

$g(x) =$ _____

4. Vertex $(3, 1)$; the graph passes through $(5, 9)$. $f(x) =$ _____

Your Turn

1. Describe the transformations of $y = x^2$: $g(x) = 3(x - 2)^2 + 1$

vertical stretch by 3, right 2, up 1

2. Describe the transformations of $y = x^2$: $g(x) = -\frac{1}{4}(x + 5)^2$

reflect in x -axis, compress by $\frac{1}{4}$, left 5

3. Write the function: $y = x^2$ reflected in the x -axis, then 6 right and 2 up.

$$g(x) = \underline{-(x - 6)^2 + 2}$$

4. Vertex $(3, 1)$; the graph passes through $(5, 9)$. $f(x) = \underline{2(x - 3)^2 + 1}$

Challenge

A parabola $f(x) = a(x - h)^2 + k$ passes through $(0, 5)$ and $(4, 5)$, and its minimum value is -3 . Find $f(x)$.

Same height at $x = 0$ and $x = 4 \Rightarrow$ axis of symmetry $x = \underline{\hspace{2cm}}$

Minimum value $-3 \Rightarrow$ vertex $\underline{\hspace{2cm}}$

$5 = a(0 - 2)^2 - 3 \Rightarrow a = \underline{\hspace{2cm}}$

$f(x) = \underline{\hspace{2cm}}$

Teacher: Key insight: two points with the same y -value are mirror images, so the axis is halfway between them.

Challenge

Challenge

A parabola $f(x) = a(x - h)^2 + k$ passes through $(0, 5)$ and $(4, 5)$, and its minimum value is -3 . Find $f(x)$.

Same height at $x = 0$ and $x = 4 \Rightarrow$ axis of symmetry $x = \underline{2}$

Minimum value $-3 \Rightarrow$ vertex $\underline{(2, -3)}$

$5 = a(0 - 2)^2 - 3 \Rightarrow a = \underline{2}$

$f(x) = \underline{2(x - 2)^2 - 3}$

Teacher: Key insight: two points with the same y -value are mirror images, so the axis is halfway between them.

Big Picture

Parent function

$y = x^2$, vertex _____, axis _____

Vertex form

$f(x) = a(x - h)^2 + k$, vertex _____

a

sign: opens up/down; $|a|$: _____

h

horizontal shift — _____ of the sign in $(x - h)$

k

vertical shift; also the _____ value

Big Picture

Parent function

$y = x^2$, vertex (0, 0), axis $x = 0$

Vertex form

$f(x) = a(x - h)^2 + k$, vertex (h, k)

a

sign: opens up/down; $|a|$: stretch or compression

h

horizontal shift — opposite of the sign in $(x - h)$

k

vertical shift; also the max or min value

Exit Ticket

For $f(x) = -3(x + 2)^2 + 7$:

1. Vertex: _____ Axis of symmetry: _____
2. Opens _____, so 7 is the _____ value. Range: _____
3. Transformations of $y = x^2$:

Homework: practice set 4-1 · Extra practice: Math Practice Generator, Chapter 4 → Quadratic Transformations

Exit Ticket

For $f(x) = -3(x + 2)^2 + 7$:

1. Vertex: $(-2, 7)$ Axis of symmetry: $x = -2$
2. Opens down, so 7 is the maximum value. Range: $y \leq 7$
3. Transformations of $y = x^2$:
reflect in x-axis, vertical stretch by 3, left 2, up 7

Homework: practice set 4-1 · Extra practice: Math Practice Generator, Chapter 4 → Quadratic Transformations